Drift-diffusion approximation when $\delta c/c \ll 1$

$$\frac{d\mathcal{R}}{dt} = V + \sqrt{2V}\eta \leftarrow \text{Gaussian noise } \langle \eta(t)\eta(t') \rangle \sim \delta(t-t')$$

V: Drift, $\sqrt{2V}\eta$: Diffusion

SPRT mean decision time (Siggia & Vergassola 2012)

$$\langle T_{\rm dec} \rangle = \frac{K}{V} \tanh\left(\frac{K}{2}\right)$$

$$V \approx \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \dot{\mathcal{F}}(c)$$

$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t} \left\langle \frac{\partial^2 \mathcal{L}(c)}{(\partial \ln c)^2} \right\rangle_c$$

$$\mathcal{L}(c;t) = \ln P(O(t)|c)$$